The investor's sole source of return is the difference between the price paid (PV) and full principal (FV) received at maturity. This type of bond is often referred to as a **zero-coupon bond** given the lack of intermediate interest cash flows, which for bonds are generally referred to as coupons.

EXAMPLE 1

Discount Bonds under Positive and Negative Interest Rates

While most governments issue fixed coupon bonds with principal paid at maturity, for many government issuers such as the United States, United Kingdom, or India, investors buy and sell individual interest or principal cash flows separated (or stripped) from these instruments as discount bonds. Consider a single principal cash flow payable in 20 years on a Republic of India government bond issued when the YTM is 6.70 percent. For purposes of this simplified example, we use annual compounding, that is, t in Equation 5 is equal to the number of years until the cash flow occurs.

1. What should an investor expect to pay for this discount bond per INR100 of principal?

   Solution:

   INR27.33
   We solve for PV given r of 6.70 percent, t = 20, and FV_{20} of INR100 using Equation 5:

   PV = INR100 / $(1 + 0.067)^{20}$ = INR27.33.

   We may also use the Microsoft Excel or Google Sheets PV function:

   `= PV (rate, nper, pmt, FV, type)`,

   where:

rate = the market discount rate per period,

nper = the number of periods,

pmt = the periodic coupon payment (zero for a discount bond),

FV = future or face value, and

type = payments made at the end (0 as in this case) or beginning (1) of each period.

   As a cash outflow (or price paid), the Excel PV solution has a negative sign, so:

   PV = (27.33) = `PV (0.067,20,0,100,0)`.

   While the principal (FV) is a constant INR100, the price (PV) changes as both time passes, and interest rates change.

2. If we assume that interest rates remain unchanged, what is the price (PV) of the bond in three years' time?

   Solution:

   INR33.21